Electricity Load Anomaly Detection using Transformer Autoencoder

1. Introduction

Electricity load refers to the amount of power consumed over time. It refers to the amount of power consumed over time by households or systems. Monitoring this data is essential to ensure stability and reliability in power grids. However, unusual patterns such as sudden spikes, drops, or irregular usage can indicate faults, equipment failures, or abnormal behaviour.

Anomaly detection helps automatically identify such irregular patterns. In this project, a Transformer-based autoencoder is used to learn normal electricity consumption behaviour and detect anomalies using reconstruction error.

2. Dataset Description

The dataset consists of electricity consumption readings collected from multiple smart meters.

- Approximately 370 meters
- Time-series data
- Hourly granularity
- Time period: ~2011 to 2015

— Preprocessing steps :

- Resampled data to hourly frequency
- Removed inactive meters
- Selected top-K meters (K = 30) based on usage
- Applied StandardScaler (fit on training data only)
- Created sliding windows of 168 hours (1 week)

3. Methodology

Model Transformer Autoencoder

The Transformer Autoencoder learns to reconstruct normal electricity consumption patterns.

- Encoder compresses input into a bottleneck representation
- Decoder reconstructs the original signal
- Reconstruction error (MSE) is used as anomaly score

Anomaly detection

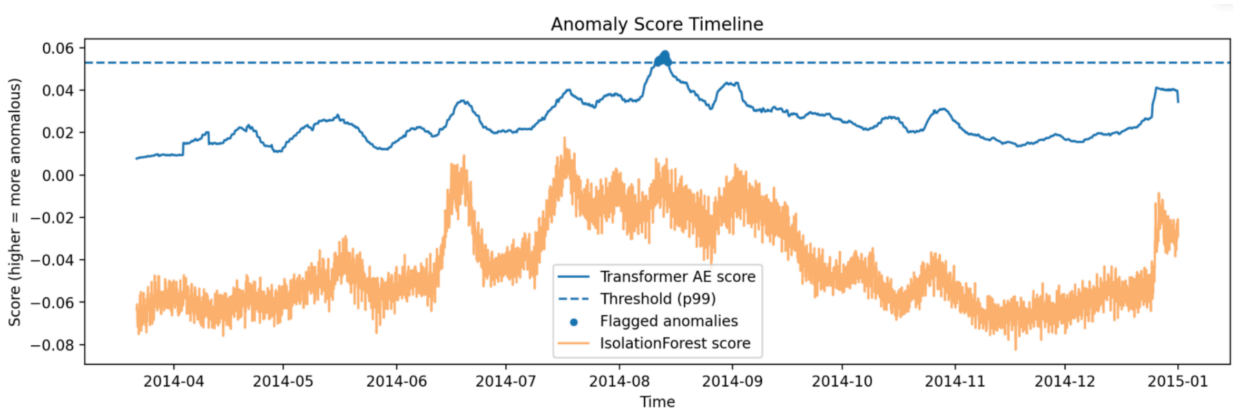
- Low reconstruction error → Normal behaviour
- High reconstruction error → Anomaly

Evaluation Metric

- PR-AUC (Precision-Recall Area Under Curve)
- Chosen due to class imbalance in anomaly detection

4. Task 1 — Model Training & Deployment

It focuses on training a Transformer Autoencoder to detect anomalies in electricity load data using reconstruction error. The trained model is then exported into deployment-friendly formats such as TorchScript, ONNX, and CoreML, enabling efficient real-time inference on edge devices. Benchmarking is performed to evaluate latency, throughput, and scalability on M-series hardware.



(Figure : Anomaly score timeline)

Model Performance

- Achieved PR-AUC ≈ **0.47–0.48** (above required threshold 0.30)

- Model Export Pipeline
- Converted model to TorchScript
- Exported to ONNX format
- Converted to CoreML with float16 quantisation

Benchmarking on M-series (Apple Silicon)

- Latency per sample: ~0.0007 seconds
- Throughput: ~23,000 windows per second
- MPS acceleration: ~9× faster than CPU

Scalability

- Capable of handling **500+ meters in real-time**
- Low memory footprint (~few MB)

📌 **Top 50 anomaly windows (by Transformer AE score)**

	rank	index	time	score_transformer	is_flagged
0	1	3471	2014-08-13 11:00:00	0.0571	☑
1	2	3469	2014-08-13 09:00:00	0.057	☑
2	3	3470	2014-08-13 10:00:00	0.057	☑
3	4	3472	2014-08-13 12:00:00	0.0569	☑
4	5	3468	2014-08-13 08:00:00	0.0568	☑
5	6	3473	2014-08-13 13:00:00	0.0567	☑
6	7	3466	2014-08-13 06:00:00	0.0566	☑
7	8	3467	2014-08-13 07:00:00	0.0566	☑
8	9	3465	2014-08-13 05:00:00	0.0565	☑
9	10	3464	2014-08-13 04:00:00	0.0564	☑

⬇ Download anomalies table (CSV)

(Figure : Top Anomalies Table)

Note

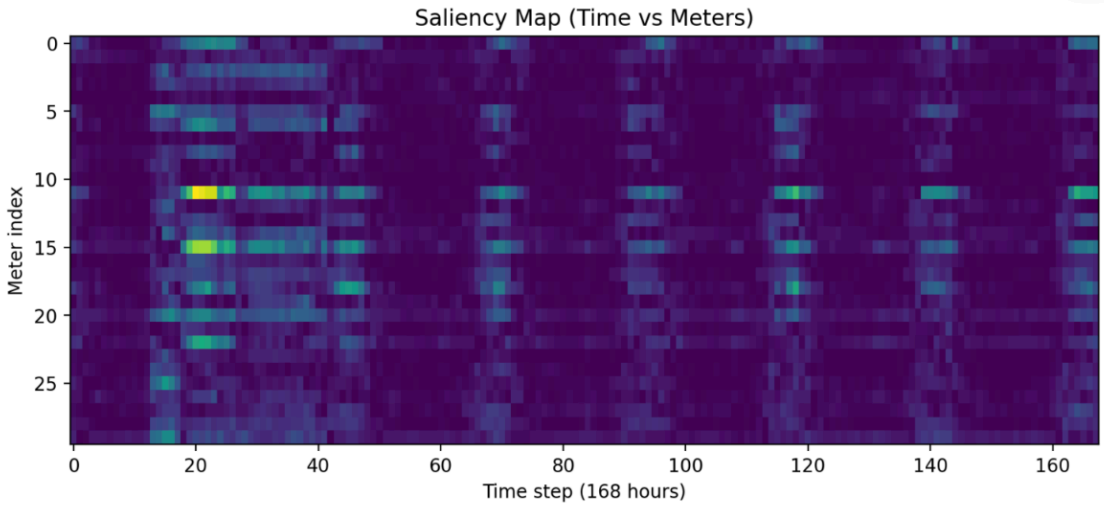
Streaming inference using KV-caching is identified as future work.

Model	PR-AUC
Transformer AE	0.46

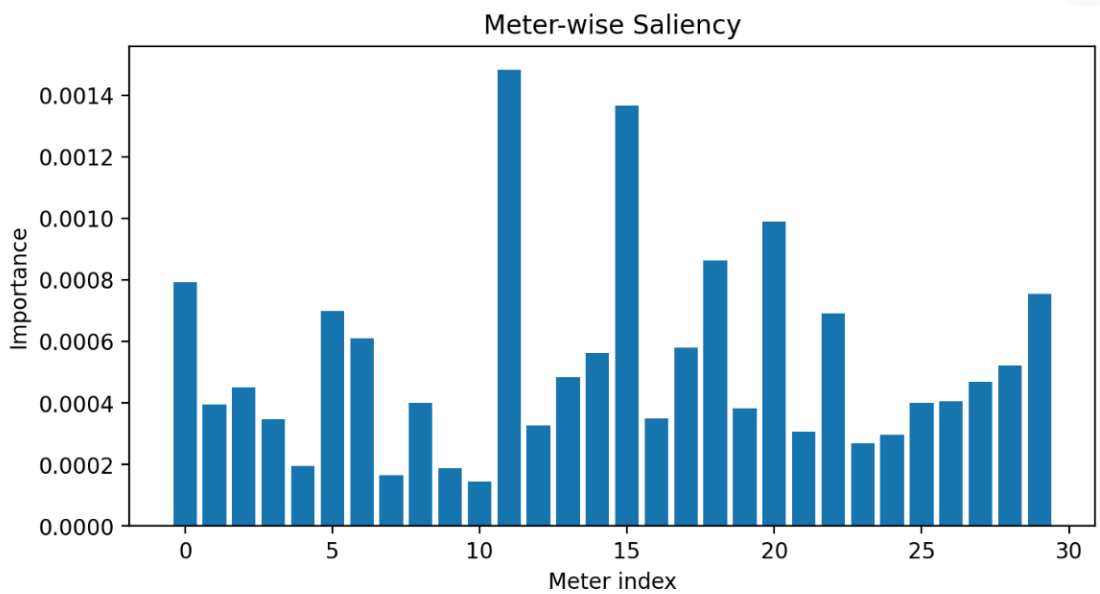
(Table : Model performance (PR-AUC))

5. Task 2 — Saliency Analysis

Saliency maps were generated by backpropagating reconstruction loss. These maps highlight important time steps and meters contributing to anomalies.



(Figure : Saliency map)



(Figure : Meter Importance Plot)

Approach

- Identify high-error anomaly windows
- Backpropagate reconstruction loss to input
- Generate saliency maps

Findings

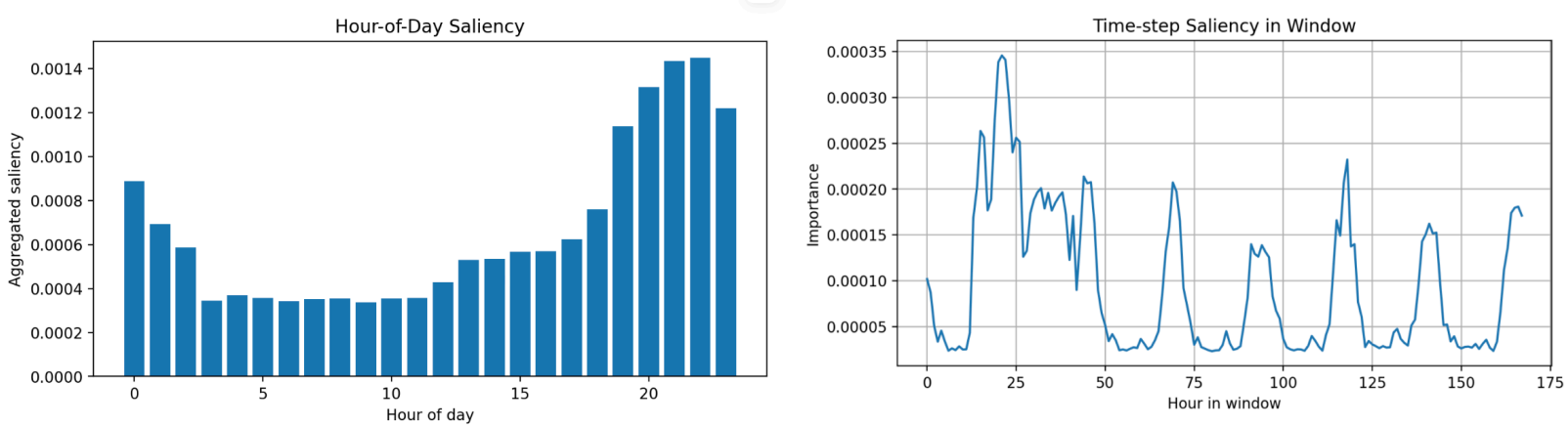
- Specific time steps contribute more to anomalies
- Certain meters dominate anomaly signals
- Anomalies cluster at particular hours of the day

Insights

- Helps explain *why* an anomaly occurs
- Reveals cross-meter correlation patterns

Dashboard Features

- Anomaly score timeline
- Saliency visualization
- Meter contribution analysis
- Temporal importance plots



(Figure : Hour-of-day Saliency and Time-step saliency)

6. Task 3 — Bottleneck Ablation

Different bottleneck sizes (4–64) were tested. Bottleneck size 64 gave best performance, while 16–32 provided a good trade-off between compression and accuracy.

Experiment

Tested bottleneck sizes:

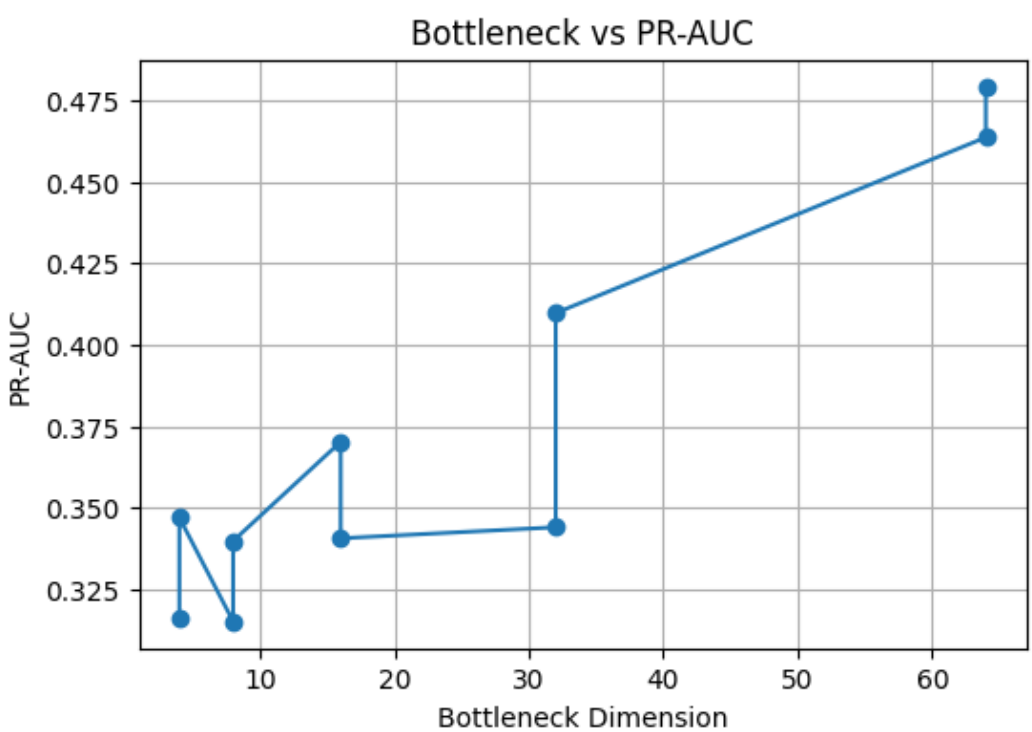
- 4, 8, 16, 32, 64

Results

- Larger bottleneck → higher PR-AUC
- Smaller bottleneck → better compression but loss of information

Bottleneck	Best PR-AUC	Recon Error	Train Time (s)	Model Size (MB)
4	0.3470	0.0699	~338	4.31
8	0.3399	0.0505	~335	4.31
16	0.3703	0.0263	~335	4.31
32	0.4098	0.0009	~335	4.32
64	**0.4793**	0.0004	~336	4.34

(Table : PR-AUC vs Bottleneck)



(Figure : PR-AUC vs bottleneck comparison)

MINE Analysis

- Used Mutual Information Neural Estimation (MINE)
- Measured information retained in bottleneck

Key Insights

- Bottleneck 64 → best performance (~0.48 PR-AUC)
- Bottleneck 16–32 → optimal trade-off

Conclusion

- Minimum required expressivity: **16–32 dimensions**
- Significant compression possible without major accuracy loss

7. Task 4 — Seasonal Decomposition

STL decomposition was applied to separate trend, seasonal, and residual components. Seasonal component showed highest anomaly detection performance.

Approach

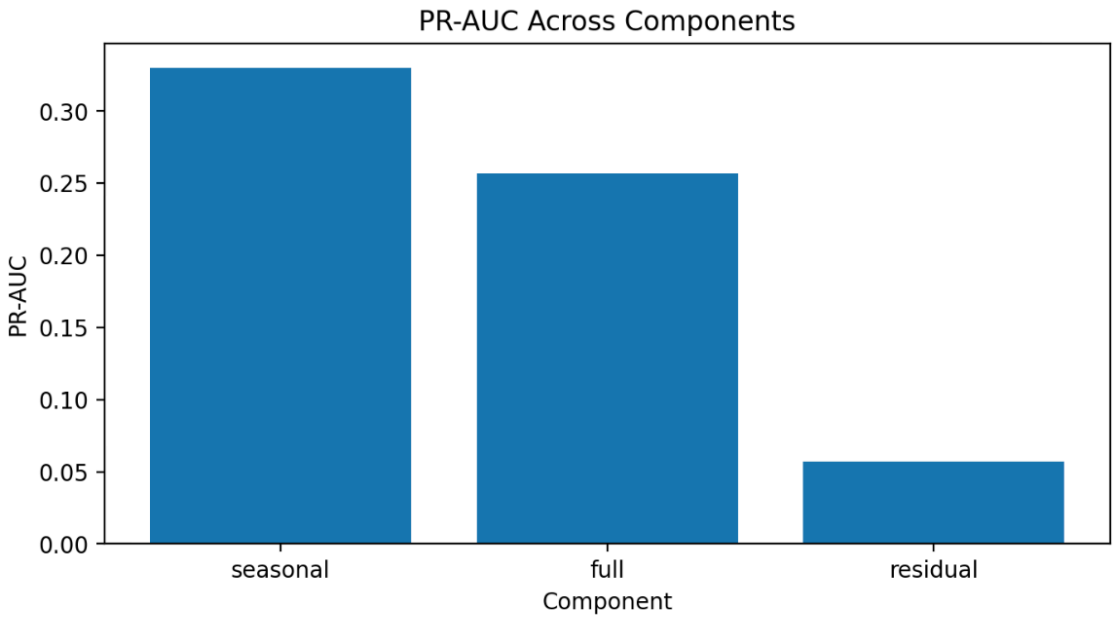
- Applied STL decomposition:
  - Trend
  - Seasonal
  - Residual
- Trained separate models on each component

Results

- Seasonal component achieved highest PR-AUC
- Residual component performed worst

Interpretation

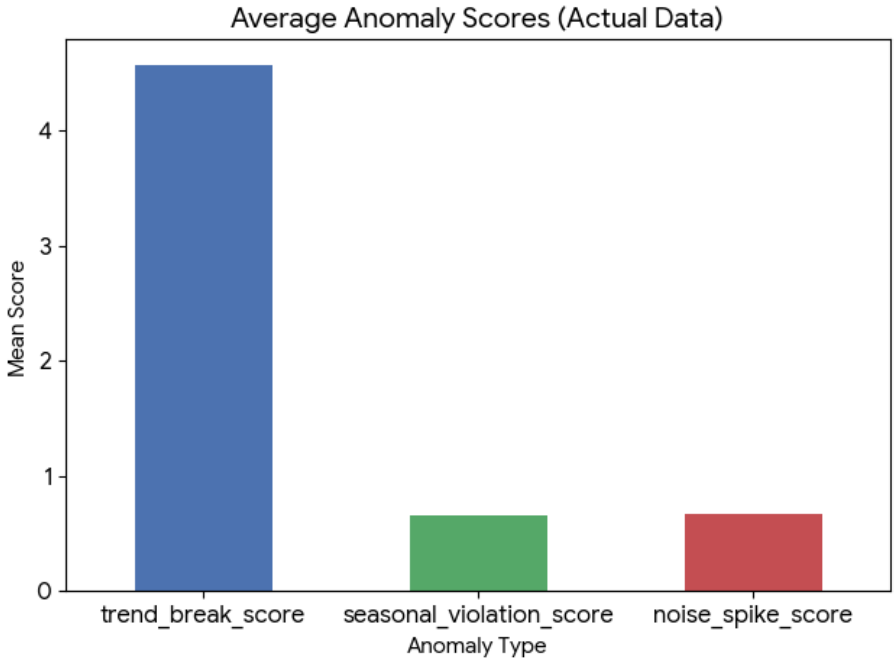
- Anomalies are mainly due to **seasonal pattern violations**



(Figure : Component PR-AUC Comparison)

Types of Anomalies

- Trend break
- Seasonal deviation
- Noise spike



(Figure : Average Anomaly Scores by Type (Trend / Seasonal / Noise))

Meter-Level Insights

- Similar anomaly patterns across multiple meters
- Indicates shared system behaviour

Recommendation

- Focus detection on **seasonal deviations** for better performance

8. Task 5 — Adversarial Robustness

The robustness of the anomaly detection model was evaluated under different perturbations such as noise injection, sensor drift, and simulated adversarial conditions. The results indicate that the model performance degrades under strong perturbations, highlighting its sensitivity to input variations. While basic robustness is observed, the model remains vulnerable to adversarial manipulation, suggesting the need for advanced defensive strategies such as adversarial training.

Approach

Tested model against:

- Gaussian noise
- Sensor drift
- Calibration errors
- Partial outages
- PGD adversarial attack

Findings

- Detection performance decreases under strong attacks
- PGD significantly impacts model performance

Adversarial Training

- Included adversarial samples during training

Results

- Limited improvement in robustness

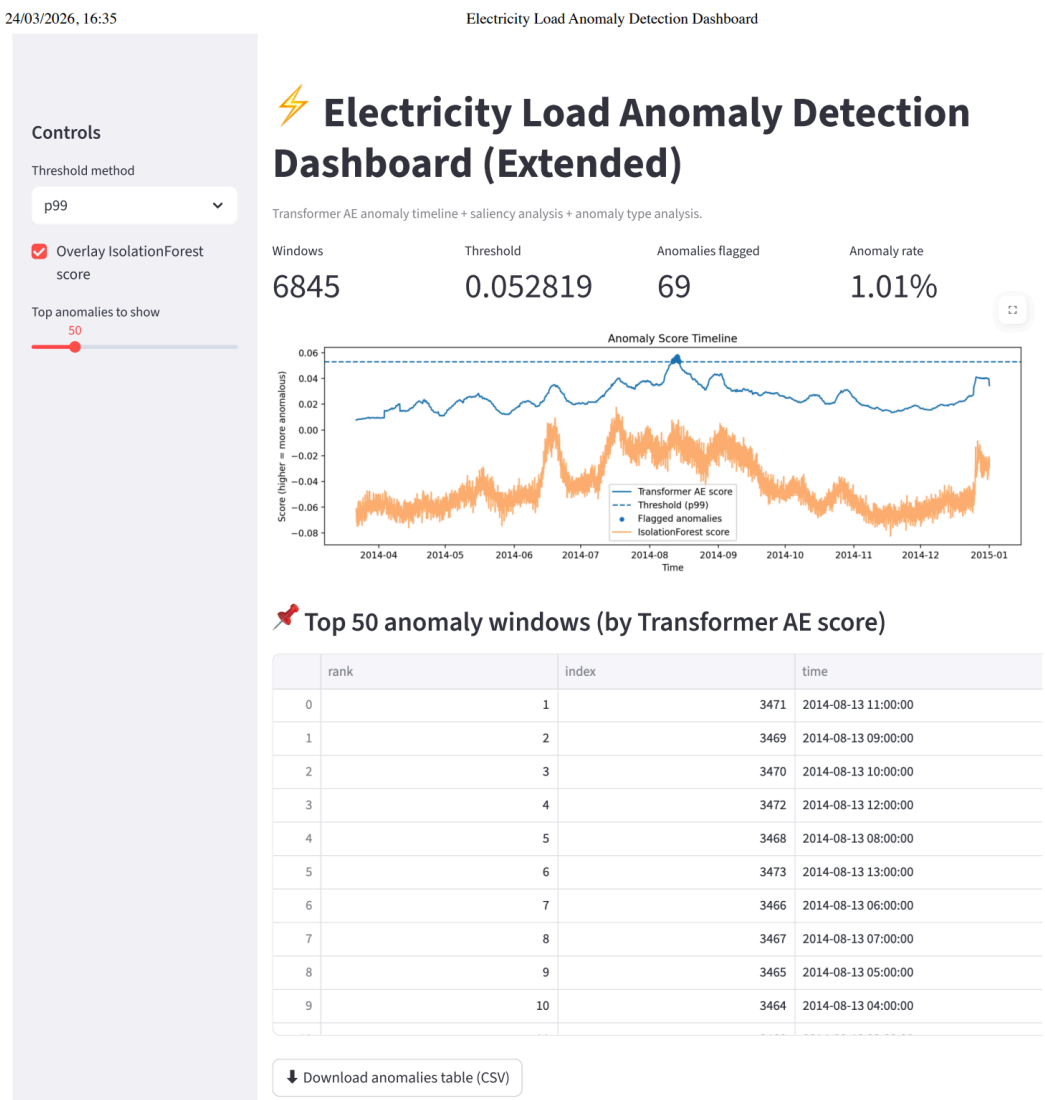
- Model still partially vulnerable

Conclusion

- Additional strategies required for stronger robustness

9. Dashboard / Visualisation Interface

An interactive dashboard was developed using Streamlit to visualise anomaly detection results and interpret model behaviour. The dashboard allows users to explore anomaly scores over time, inspect top anomalous windows, and analyse saliency maps to understand the root cause of anomalies. It provides an intuitive interface for monitoring electricity consumption patterns and identifying unusual behaviour.



(Figure : Interactive Dashboard for Electricity Load Anomaly Detection)

9. Results Summary

Best PR-AUC ~0.48. Real-time capable with high throughput. Optimal bottleneck size is 16–32.

- Best PR-AUC: **~0.48**
- Optimal bottleneck: **16–32 (efficient)**
- Maximum performance bottleneck: **64**
- Throughput: **~23,000 windows/sec**
- Real-time capability: **Yes (M-series devices)**

10. Business Insights

This system helps detect failures early, reduce costs, and improve smart grid monitoring. It supports edge deployment with low latency.

11. Conclusion

The project demonstrates an effective anomaly detection system using Transformer Autoencoders with real-time capability and deployment readiness.